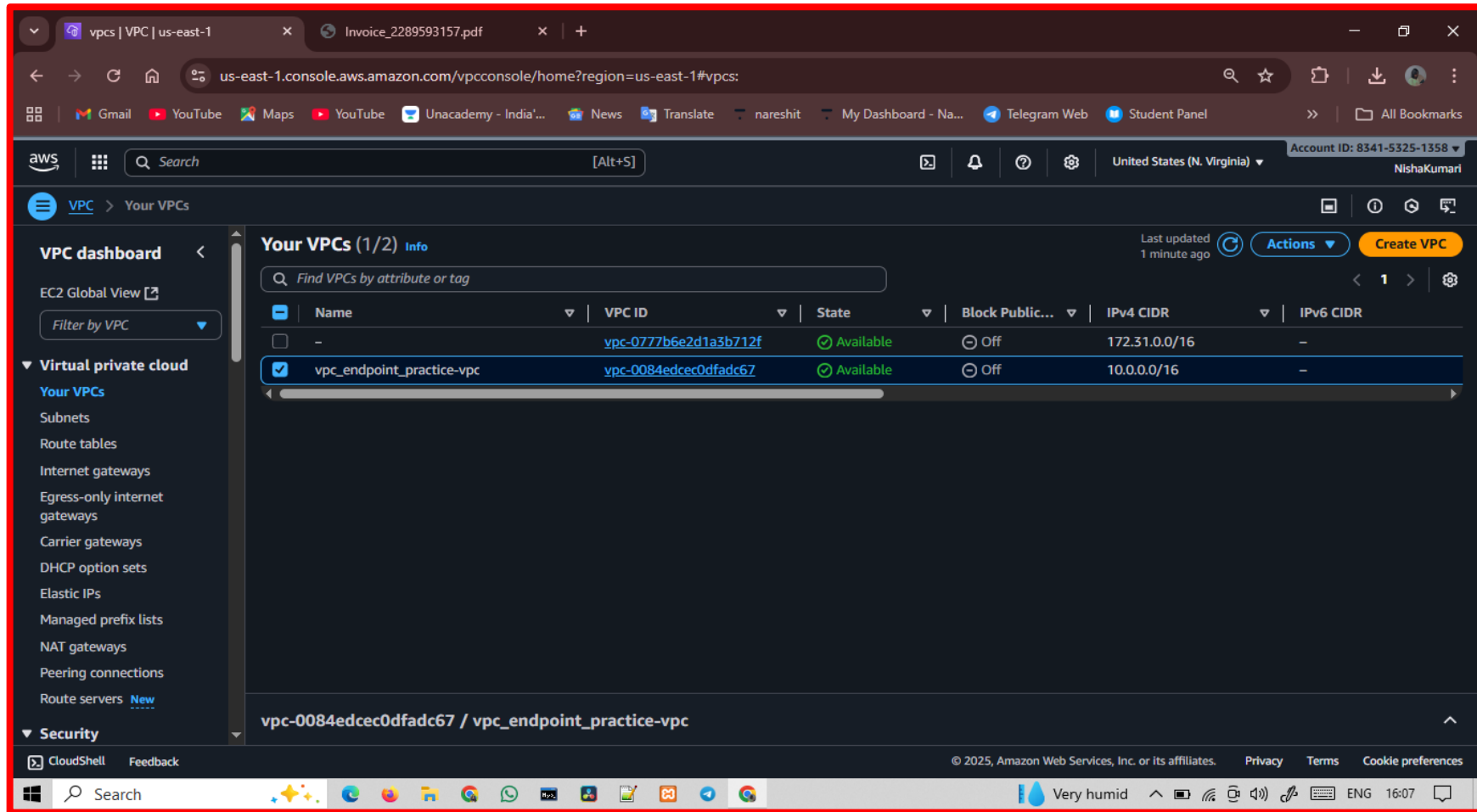


DATE : 23/SEP/2025
TOPIC : VPC ENDPOINT
TEAM : SKYOPS CREW

Step 1: create vpc and subnet ..



The screenshot displays the AWS Management Console interface for the 'us-east-1' region. The left-hand navigation pane shows the 'VPC dashboard' with a list of VPC-related services. The main content area, titled 'Your VPCs (1/2)', contains a table of existing VPCs. Two VPCs are listed: 'vpc-0777b6e2d1a3b712f' and 'vpc_endpoint_practice-vpc'. The second VPC is selected, and its details are shown in the bottom section of the console.

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
-	vpc-0777b6e2d1a3b712f	Available	Off	172.31.0.0/16	-
☒ vpc_endpoint_practice-vpc	vpc-0084edcec0dfadc67	Available	Off	10.0.0.0/16	-

Below the table, the details for the selected VPC 'vpc-0084edcec0dfadc67 / vpc_endpoint_practice-vpc' are displayed.

Step 2 : launch 2 instances

i) Private instance

ii) Public instance

Public instance in public subnet

The screenshot displays the AWS Management Console interface for an EC2 instance. The browser tabs at the top include 'subnets | VPC | us-east-1', 'Instance details | EC2 | us-east-1', and 'Invoice_2289593157.pdf'. The address bar shows the URL 'us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-061fc287678a654bf'. The console header shows the AWS logo, a search bar, and the account ID '8341-5325-1358' for 'NishaKumari' in the 'United States (N. Virginia)' region.

The main content area is titled 'Instance summary for i-061fc287678a654bf (public-server)' and is updated 'less than a minute ago'. It features a 'Connect' button, an 'Instance state' dropdown menu, and an 'Actions' dropdown menu.

Instance ID	Public IPv4 address	Private IPv4 addresses
i-061fc287678a654bf	54.160.211.231 open address	10.0.10.200

IPV6 address	Instance state	Public DNS
-	Running	ec2-54-160-211-231.compute-1.amazonaws.com open address

Hostname type	Private IP DNS name (IPv4 only)	Elastic IP addresses
IP name: ip-10-0-10-200.ec2.internal	ip-10-0-10-200.ec2.internal	-

Answer private resource DNS name	Instance type	AWS Compute Optimizer finding
-	t3.micro	Opt-in to AWS Compute Optimizer for recommendations. Learn more

Auto-assigned IP address	VPC ID	Auto Scaling Group name
54.160.211.231 [Public IP]	vpc-0084edcec0dfadc67 (vpc_endpoint_practice-vpc)	-

IAM Role	Subnet ID	Managed
-	subnet-08bbcf85d6bfec2f6 (vpc_endpoint_practice-subnet-public1-us-east-1a)	false

IMDSv2	Instance ARN
Required	arn:aws:ec2:us-east-1:834153251358:instance/i-061fc287678a654bf

The bottom of the console shows a 'CloudShell' button, copyright information for Amazon Web Services, and links for 'Privacy', 'Terms', and 'Cookie preferences'. The Windows taskbar at the very bottom shows the search bar, various application icons, and system status information including '29°C Cloudy' and the time '16:23'.

Private instance in private subnet:-

The screenshot displays the AWS Management Console interface for an EC2 instance. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0e206a024efd903cd`. The console header includes the AWS logo, a search bar, and navigation links for EC2, Instances, and the specific instance ID `i-0e206a024efd903cd`. The instance summary for `i-0e206a024efd903cd (private-server)` is shown, indicating it is in the `Running` state. The instance is configured with a private IP address of `10.0.135.106` and is associated with the `subnet-05ca2530457a6d220` subnet. The instance type is `t3.micro` and the VPC ID is `vpc-0084edcec0dfadc67`. The instance is not associated with a public IP address or an Elastic IP address. The IAM Role is `None` and the IMDSv2 setting is `Required`. The instance is managed by AWS and is not part of an Auto Scaling Group.

Property	Value
Instance ID	<code>i-0e206a024efd903cd</code>
IPv6 address	None
Hostname type	IP name: <code>ip-10-0-135-106.ec2.internal</code>
Answer private resource DNS name	None
Auto-assigned IP address	<code>18.215.178.146</code> [Public IP]
IAM Role	None
IMDSv2	Required
Operator	None
Public IPv4 address	<code>18.215.178.146</code> open address
Instance state	Running
Private IP DNS name (IPv4 only)	<code>ip-10-0-135-106.ec2.internal</code>
Instance type	<code>t3.micro</code>
VPC ID	<code>vpc-0084edcec0dfadc67</code> (vpc_endpoint_practice-vpc)
Subnet ID	<code>subnet-05ca2530457a6d220</code>
Instance ARN	<code>arn:aws:ec2:us-east-1:834153251358:instance/i-0e206a024efd903cd</code>
Private IPv4 addresses	<code>10.0.135.106</code>
Public DNS	<code>ec2-18-215-178-146.compute-1.amazonaws.com</code> open address
Elastic IP addresses	None
AWS Compute Optimizer finding	Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto Scaling Group name	None
Managed	false

Step 3 : edit inbound rules of security group :-

The screenshot shows the AWS Management Console interface for editing inbound rules of a security group. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-080d9fdc34a0ce4ce`.

The console header shows the AWS logo, a search bar, and the account ID: 8341-5325-1358. The breadcrumb navigation is: `EC2 > Security Groups > sg-080d9fdc34a0ce4ce - default > Edit inbound rules`.

The main section is titled "Edit inbound rules" with an "Info" link. Below the title, it states: "Inbound rules control the incoming traffic that's allowed to reach the instance."

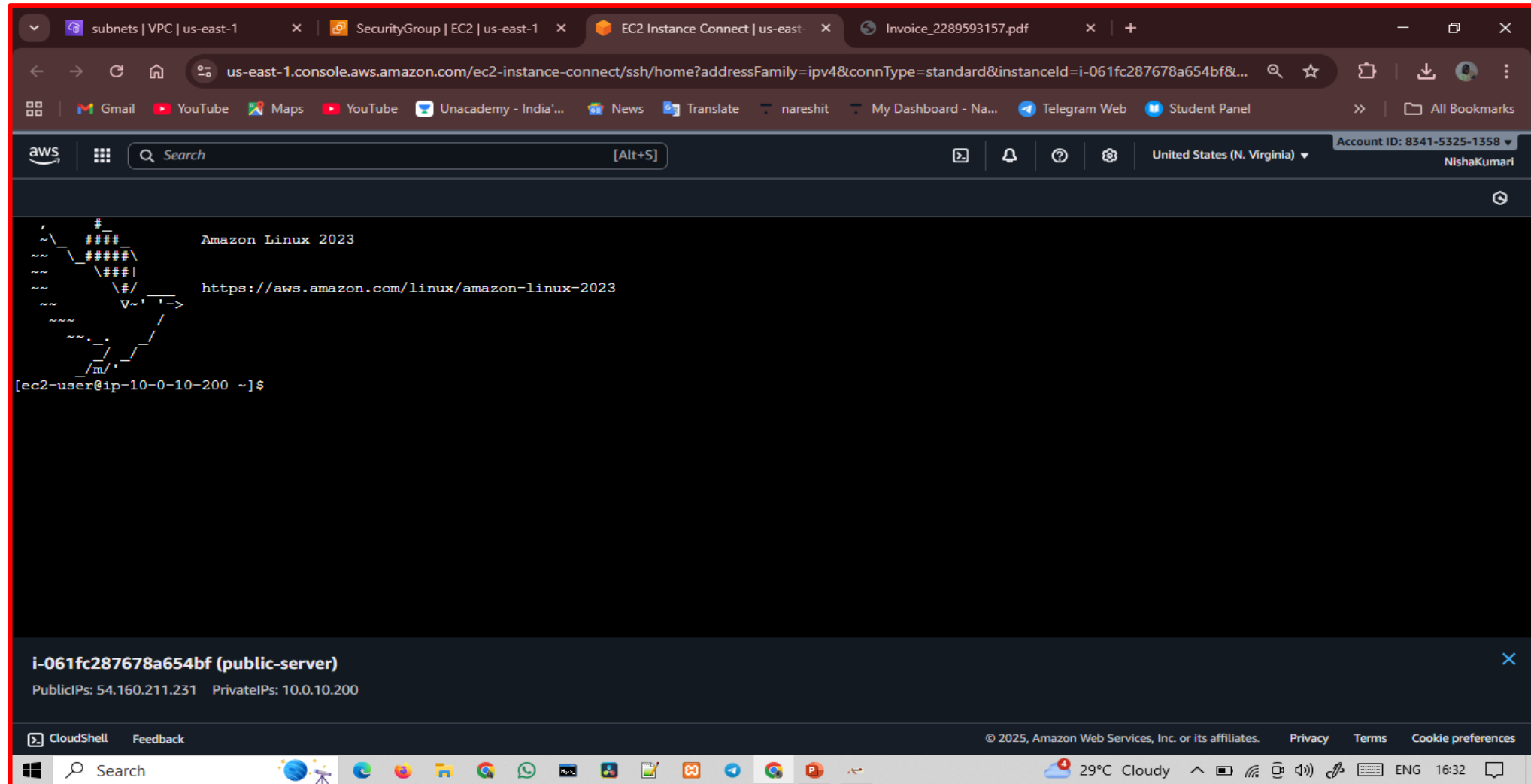
The "Inbound rules" section contains a table with the following columns: Security group rule ID, Type, Protocol, Port range, Source, and Description - optional. There are two existing rules and one rule being added.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-0b033b5fe762fa483	All traffic	All	All	Custom	
-	All traffic	All	All	Anyw...	
				0.0.0.0/0	

At the bottom of the console, there are buttons for "Add rule", "Cancel", "Preview changes", and "Save rules". The "Save rules" button is highlighted in yellow.

The footer of the console shows the CloudShell logo, copyright information (© 2025, Amazon Web Services, Inc. or its affiliates.), and links for Privacy, Terms, and Cookie preferences. The system tray at the bottom of the screen shows the date and time (16:30) and the temperature (29°C Cloudy).

Step 4: connect public instance to public server



Step 5 : run command

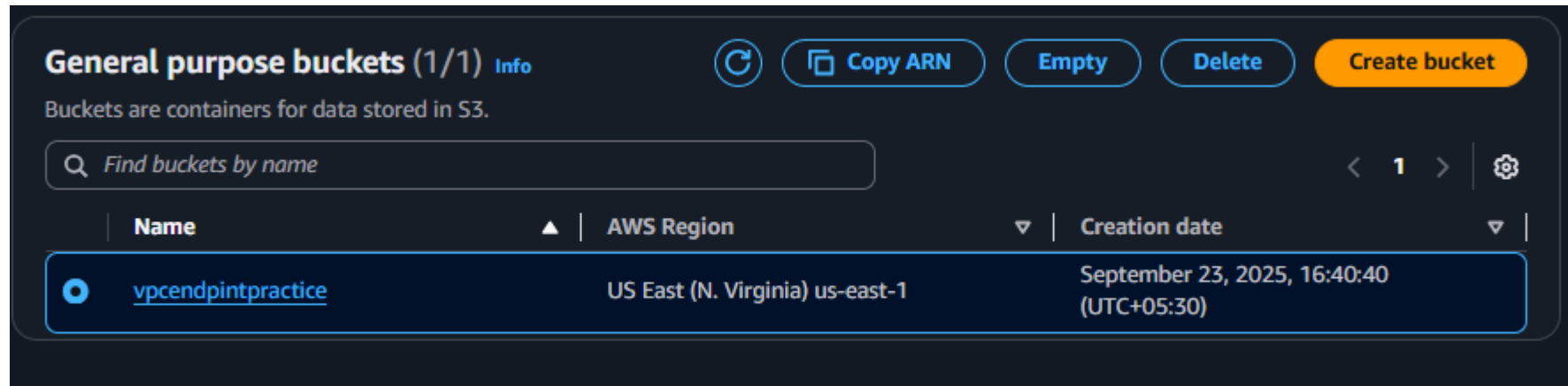
i)sudo su – [to convert to root user]

ii)aws s3 ls [try to see list of s3 bucket and error {otput}]

```
/ m/  
[ec2-user@ip-10-0-10-200 ~]$ sudo su -  
[root@ip-10-0-10-200 ~]# aws s3 ls  
  
Unable to locate credentials. You can configure credentials by running "aws configure".  
[root@ip-10-0-10-200 ~]# █
```

Step 6 :- create s3 bucket and again try to view bucket in aws consol using command {aws s3 ls}

Creation of s3 bucket :



Output :-

```
[root@ip-10-0-10-200 ~]# aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws configure".
[root@ip-10-0-10-200 ~]#
```

Step 7 :- create IAM role give s3 permission

The screenshot displays the AWS IAM console interface. The left sidebar shows the navigation menu with 'IAM' selected. The main content area shows the details for the 'vpcentpoint-iam' role. The 'Summary' section includes the creation date (September 23, 2025, 16:48 UTC+05:30), the role's ARN, and the instance profile ARN. The 'Permissions' tab is active, showing a table of attached policies. One policy, 'AmazonS3FullAccess', is listed with the type 'AWS managed' and is attached to 1 entity. The 'Permissions boundary' section indicates that no boundary is set.

vpcentpoint-iam Info
Allows EC2 instances to call AWS services on your behalf.

Summary

- Creation date: September 23, 2025, 16:48 (UTC+05:30)
- ARN: `arn:aws:iam::834153251358:role/vpcentpoint-iam`
- Instance profile ARN: `arn:aws:iam::834153251358:instance-profile/vpcentpoint-iam`
- Last activity: -
- Maximum session duration: 1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (1) Info

You can attach up to 10 managed policies.

Policy name	Type	Attached entities
AmazonS3FullAccess	AWS managed	1

Permissions boundary (not set)

Step 8:-connect private instance to server

```
[root@ip-10-0-10-200 ~]# vi vpc-endpoint.pem
[root@ip-10-0-10-200 ~]# chmod 400 "vpc-endpoint.pem"
[root@ip-10-0-10-200 ~]# ssh -i "vpc-endpoint.pem" ec2-user@ec2-18-215-178-146.compute-1.amazonaws.com

      #
     _###_      Amazon Linux 2023
    ~\  _###_#
   ~ ~ \_#####\
   ~ ~  \###|
   ~ ~   \#/      https://aws.amazon.com/linux/amazon-linux-2023
   ~ ~    V~'  ' ~>
      ~ ~ ~
      ~ ~ . .
      ~ ~ / /
      ~ ~ /m/ '

[ec2-user@ip-10-0-135-106 ~]$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-10-0-135-106 ~]$
```

Step 9: go to private instance and attach iam role for private instance

This screenshot shows the AWS Management Console 'Instance details' page for a private EC2 instance. The instance is named 'i-0e206a024efd903cd' and is in a 'Running' state. The left sidebar shows the navigation menu with 'Instances' selected. The main content area displays various instance attributes in a grid:

- Instance ID:** i-0e206a024efd903cd
- Public IPv4 address:** 18.215.178.146
- Private IPv4 address:** 10.0.135.106
- Instance state:** Running
- Hostname type:** IP name: ip-10-0-135-106.ec2.internal
- Answer private resource DNS name:** -
- Auto-assigned IP address:** 18.215.178.146 [Public IP]
- IAM Role:** -
- IMDSv2:** Required
- Instance type:** t3.micro
- Private DNS name (IPv4 only):** ip-10-0-135-106.ec2.internal
- VPC ID:** vpc-0084edec0dfad67 (vpc_endpoint_practice-vpc)
- Subnet ID:** subnet-05ca2530457a6d220 (vpc_endpoint_practice-subnet-private1-us-east-1a)
- Instance ARN:** arn:aws:ec2:us-east-1:834153251358:instance/i-0e206

A context menu is open over the 'IAM Role' field, showing options: 'Change security groups', 'Get Windows password', and 'Modify IAM role'. The 'Modify IAM role' option is highlighted.

This screenshot shows the 'Modify IAM role' page in the AWS Management Console. The page title is 'Modify IAM role' and it includes a sub-header 'Attach an IAM role to your instance.' The main content area shows the 'Instance ID' as 'i-0e206a024efd903cd (private-server)' and the 'IAM role' as 'vpcendpoint-iam'. There is a 'Create new IAM role' button next to the role name. At the bottom right, there are 'Cancel' and 'Update IAM role' buttons.

Step 10: create endpoint

✔ Successfully created VPC endpoint
vpce-0018e99783fe0a1da

Details

Endpoint ID

vpce-0018e99783fe0a1da

VPC ID

vpce-0084edcec0dfadc67
(vpc_endpoint_practice-vpc)

Service region

us-east-1

Status

✔ Available

Status message

—

Creation time

Tuesday, September 23, 2025 at 17:12:29 GMT+5:30

Service name

com.amazonaws.us-east-1.s3

Endpoint type

Gateway

Private DNS names enabled

No

Route tables | Policy | Tags

Route tables

🔍 Search

Manage route tables

< 1 > ⚙️

Name	Route Table ID	Main	Associated Id
No route tables attached to endpoint			

Step 11: modify routable and try to access s3 bucket by command{aws s3 ls}

The screenshot shows the AWS Management Console for a VPC endpoint. The breadcrumb navigation is VPC > Endpoints > vpc-0018e99783fe0a1da. The left sidebar shows the navigation menu with categories like Getting started, DNS firewall, and Network Firewall. The main content area displays the details of the VPC endpoint, including its ID, VPC ID, service region, creation time, and status (Available). Below the details, the 'Route tables' tab is selected, showing a table with 3 route tables. The table has columns for Name, Route Table ID, Main, and Associated Id.

Name	Route Table ID	Main	Associated Id
vpc_endpoint_practice-rtb-private1-us-ea...	rtb-07ad851502629d24c (vpc_endpoin...	No	subnet-05ca2530457a6d220 (vpc_end...
-	rtb-0f538d3eace3c3e2f	Yes	-
vpc_endpoint_practice-rtb-public	rtb-0098698401e389dc6 (vpc_endpoin...	No	subnet-08bcbf85d6bfec2f6 (vpc_endp...

```
[ec2-user@ip-10-0-135-106 ~]$ aws s3 ls
2025-09-23 11:10:40 vpcendpintpractice
[ec2-user@ip-10-0-135-106 ~]$
```